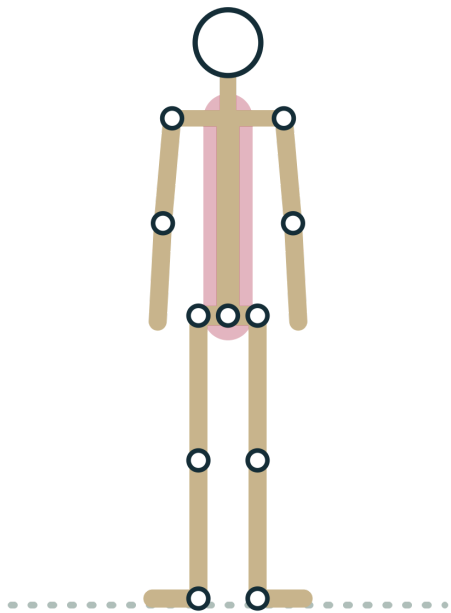


Module 1

The Language of Movement

A shared vocabulary of planes, axes and joint movements lets you cue any pose clearly and understand every module that follows.



1.5 hours • 4 lessons • Dr. Akash Garg, D.C.
Anatomy & Physiology for Vinyasa Teachers

Overview

Before you can explain why a pose works, you need words that mean the same thing to everyone in the room. This module builds that vocabulary: anatomical position, directional terms, the three planes of the body with their axes, and the paired terms for joint movement, from flexion and extension to the special words for the ankle, forearm and shoulder blade.

We learn it all through three standing poses you already know: Mountain, Warrior II and Triangle. You will also meet the six tissues that every pose loads, with a clear line between what research supports and what remains yoga lore. Throughout, the rule for teachers is simple: adapt and refer, never diagnose.

By the end of this module you can

1. Describe anatomical position and explain how it differs from Mountain pose
2. Identify the sagittal, frontal and transverse planes and the axis of each in Mountain, Warrior II and Triangle
3. Name the joint movements of a standing pose, joint by joint, using correct paired terms
4. Define the special movement terms for the ankle, foot, forearm and shoulder blade and give a pose example of each
5. Compare bone, cartilage, ligament, tendon, fascia and muscle by function and ability to adapt
6. Distinguish well supported anatomy claims from common yoga myths

Lessons

- **1.1** Why anatomy makes a better yoga teacher (15 min)
- **1.2** Anatomical position, planes and axes, seen through three standing poses (25 min)
- **1.3** Movement vocabulary: flexion, extension, abduction, adduction, rotation and special terms (30 min)
- **1.4** Tissues in brief: bone, cartilage, ligament, tendon, fascia, muscle (20 min)

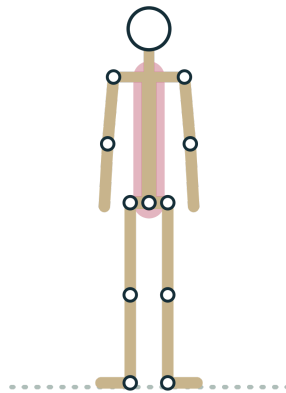
1.1 Why anatomy makes a better yoga teacher

Anatomy makes you a better yoga teacher for three practical reasons. First, it gives you precise language. When you say lift your arms in the frontal plane or bend your front knee, every student hears the same instruction. Second, it helps you understand variation. Bones, joints and connective tissue differ from person to person, so the same pose will look and feel different in different bodies. Knowing why lets you offer options instead of forcing one shape. Third, it helps you recognise when something is beyond a yoga class, so you can refer a student to a qualified clinician.

This course uses a four step method in every lesson. See it: look at the structure. Feel it: find it in your own body on the mat. Cue it: turn it into plain words for class. Check it: answer a few questions to confirm you understand. The method mirrors how adults learn movement best, by linking a concept to a physical sensation and then using it.

You will also learn to separate evidence from lore. Many popular claims, such as twists detoxifying organs or stretching releasing emotions stored in fascia, are not supported by research. Other benefits, such as improved balance, strength and flexibility, and the calming effect of slow breathing for many people, are reasonably well supported. Saying plainly what we know and what we do not protects students and builds your credibility.

Finally, remember your scope. Yoga teachers adapt and refer; they never diagnose. If a student reports sharp pain, pain that persists, numbness or tingling, offer a comfortable alternative and suggest they see a doctor or physical therapist.



Even Mountain pose is full of anatomy worth naming

KEY POINTS

- Anatomy gives you precise, shared language for cues
- Bodies vary, so poses should be adapted rather than forced into one shape
- Every lesson follows See it, Feel it, Cue it, Check it
- Separate well supported benefits from popular myths
- Yoga teachers adapt and refer; they never diagnose

FEEL IT

Stand in Mountain pose with your eyes open. Sway very slowly forward and back, then side to side, for about thirty seconds each. Notice where pressure moves under your feet and which areas of your body start working. Come to stillness and write down three sensations in plain words before you learn any technical terms for them.

CUE IT

"Let's find a version of this pose that feels steady for you today."

"If anything feels sharp or pinchy, ease out and take Child's pose."

"Notice where the weight sits in your feet before you change anything."

CHECK IT

A student says her knee hurts in Warrior II. What is the best first response from a yoga teacher?

Answer: Offer a modified version or rest, and suggest she see a clinician if pain persists. Teachers adapt and refer; they do not diagnose or encourage working through pain.

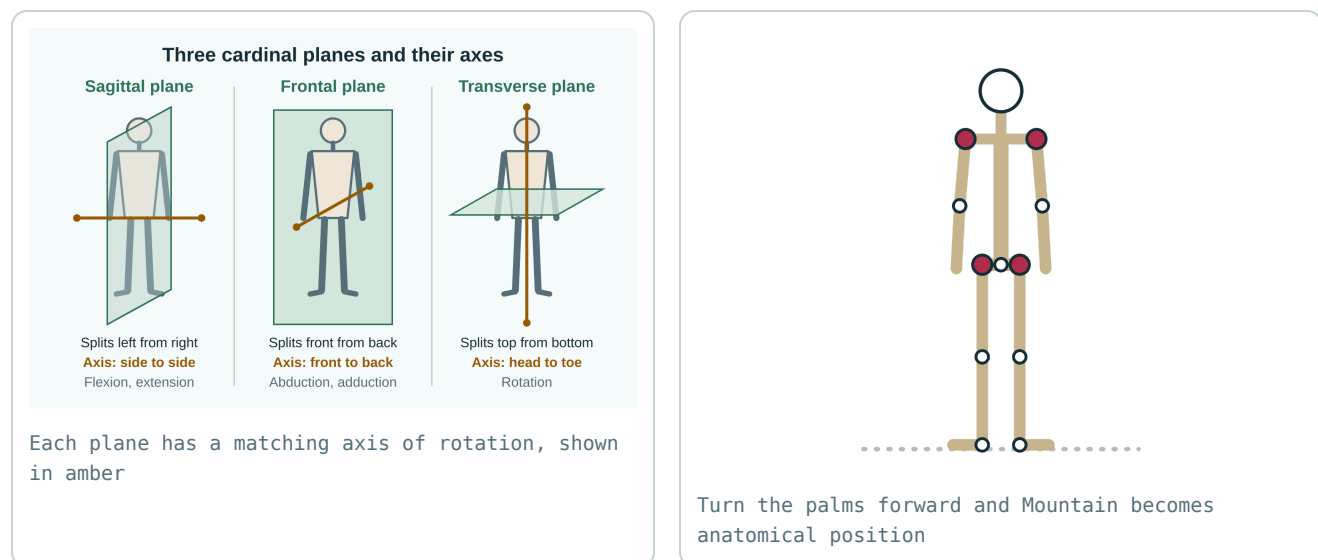
1.2 Anatomical position, planes and axes, seen through three standing poses

Anatomical position is the reference posture that every movement term refers to: standing upright, face and feet forward, arms by the sides with palms facing forward. Mountain pose (Tadasana) is almost the same, except that most students rest their palms against their thighs. Because all terms are named from this position, a word like flexion means the same thing whether a student is standing, lying down or upside down.

Directional terms describe where one structure is relative to another. They come in pairs: superior and inferior (toward the head or feet), anterior and posterior (front or back), medial and lateral (toward or away from the midline) and proximal and distal (closer to or farther from where a limb joins the trunk).

The body has three cardinal planes. The sagittal plane divides left from right, and movement in it goes forward and back, as in a forward fold. The frontal or coronal plane divides front from back, and movement in it goes out to the side and back, as in lifting the arms in Warrior II. The transverse plane divides top from bottom, and movement in it is rotation, as in a twist. Each movement turns around an axis that sits at right angles to its plane: a side to side axis for sagittal movement, a front to back axis for frontal movement and a vertical axis for transverse movement. Textbooks name these axes in slightly different ways, so focus on the idea of the hinge.

Real poses combine planes. Warrior II lifts the arms in the frontal plane, turns the front thigh out in the transverse plane and bends the front knee. Triangle tilts the trunk mostly in the frontal plane while the front hip flexes. Planes are a naming tool, not a demand that poses stay perfectly flat.



KEY POINTS

- Anatomical position is Mountain with palms turned forward
- Directional terms come in opposite pairs
- Sagittal: forward and back; frontal: side to side; transverse: turning
- Each plane has an axis at right angles to it, like a hinge
- Most poses combine movements in more than one plane

FEEL IT

Stand in Mountain and turn your palms forward to find anatomical position. Fold forward a few inches and return, noticing the sagittal plane. Lift your arms out to the sides for the frontal plane, then turn your chest gently left and right for the transverse plane. Finish in Warrior II and name the plane of each joint as you scan from arms to feet.

CUE IT

“Turn your palms forward and feel your collarbones widen.”

“Lift your arms out to the sides, in line with your shoulders.”

“Tilt sideways from the hips and keep both sides of your waist long.”

CHECK IT

Lifting the arms out to the sides into Warrior II arms happens mainly in which plane?

Answer: Frontal plane. Moving the arms out to the side and back in is abduction and adduction in the frontal plane.

1.3 Movement vocabulary: flexion, extension, abduction, adduction, rotation and special terms

Movement terms describe what a joint is doing, always relative to anatomical position, and most come in opposite pairs. Flexion usually decreases the angle between two bones, such as bending the knees in Chair, and extension increases it, such as straightening to Mountain. Abduction moves a limb away from the midline and adduction brings it back. Internal (medial) rotation turns the front of a limb toward the midline, and external (lateral) rotation turns it away. Always name the joint with the movement: hip flexion, shoulder abduction, knee extension.

Movements belong to planes. Flexion and extension, and dorsiflexion and plantarflexion, are sagittal. Abduction, adduction and lateral flexion of the spine are frontal. Rotations and horizontal abduction and adduction are transverse. Circumduction combines several movements into a circle, such as arm circles.

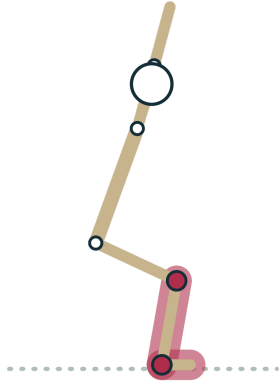
Special terms apply to particular joints. At the ankle, dorsiflexion brings the top of the foot toward the shin (Chair) and plantarflexion points the foot (tops of feet down in Cobra). Inversion turns the sole inward and eversion turns it outward (the small wobbles of the standing foot in Tree). At the forearm, supination turns the palm up and pronation turns it down (hands on the mat in Downward Dog). The shoulder blade elevates and depresses, protracts and retracts, and rotates upward and downward. Upward rotation is a normal and necessary part of lifting the arms overhead, so cues that pin the shoulders down and back with arms overhead can work against healthy shoulder mechanics.

The most practical skill is describing a pose joint by joint. In Warrior II the front hip is flexed, abducted and externally rotated, the front knee is flexed, the front ankle is dorsiflexed, the back knee is extended, and the shoulders are abducted. Practise this scan on every pose you teach.

Movement terms come in pairs

Flexion Extension Knees bend in Chair, straighten in Mountain		Abduction Adduction Arms lift out to the sides in Warrior II	
Internal rotation External rotation Hip, seen from above: Warrior II front leg		Dorsiflexion Plantarflexion Shins forward in Chair, feet pointed in Cobra	
Inversion Eversion Back view of the ankle: the wobble in Tree		Elevation Depression Shoulders creep up, then settle down	

Paired movement terms, with the first term in red and its partner in green



Chair asks for plenty of ankle dorsiflexion

KEY POINTS

- Name the joint and the movement together, such as hip flexion
- Most movement terms come in opposite pairs
- Sagittal: flexion and extension; frontal: abduction and adduction; transverse: rotation
- Ankle, foot, forearm and shoulder blade each have special terms
- Upward rotation of the shoulder blades is needed for arms overhead
- Describe poses joint by joint, from hips to shoulders

FEEL IT

Come into Warrior II on your right side. Starting at the front hip and working down to the ankle, say out loud what each joint is doing. Then do the same for the back leg and both shoulders. Change sides and repeat without looking at your notes. Finish by lifting your arms overhead and feeling your shoulder blades rotate upward.

CUE IT

“Bend your front knee and let the shin lean forward over the ankle.”

“Open your arms out to the sides at shoulder height.”

“As your arms rise overhead, let your shoulder blades glide up and out.”

CHECK IT

In Chair pose, the shins lean forward over the feet. What is the ankle doing?

Answer: Dorsiflexion. The top of the foot and the shin come closer together, which is dorsiflexion.

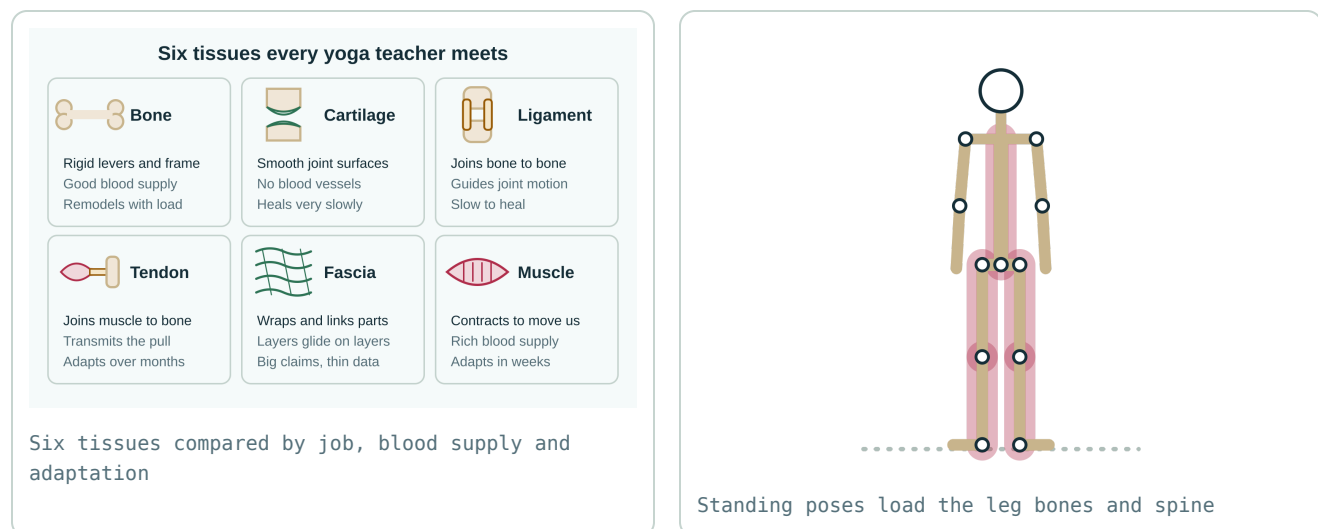
1.4 Tissues in brief: bone, cartilage, ligament, tendon, fascia, muscle

Six tissues share the work of every pose. Bone forms the rigid frame and levers of the body. The adult skeleton has 206 bones. Bone is living tissue with a blood supply and nerves; it stores minerals, makes blood cells in its marrow and remodels in response to the loads placed on it.

Cartilage covers the ends of bones inside synovial joints, giving a smooth surface that reduces friction. Articular cartilage has no blood vessels and gets nutrients by diffusion from joint fluid, which movement and loading help circulate, so it heals very slowly. Ligaments are dense bands of collagen that join bone to bone, guide and limit joint movement and contain nerve endings that sense joint position. With a limited blood supply, an overstretched ligament heals slowly and may remain loose, which is why yoga aims to stretch muscles rather than hang in the end range of joints. Tendons join muscle to bone and transmit muscle force. They strengthen with gradual loading but adapt more slowly than muscle.

Muscle is the contractile tissue. Skeletal muscle has a rich blood supply and adapts to training within weeks. Fascia is fibrous connective tissue that wraps muscles, groups of muscles and organs, linking them into a continuous web. It is richly supplied with sensory nerves. Research suggests that dense fascia requires very large forces to deform permanently, so stretching and massage probably work mainly through the nervous system rather than by quickly remodelling fascia. Claims that fascia stores emotions or toxins are not supported by evidence.

For teaching, the practical lesson is about pace and sensation. Muscles respond quickly, tendons and bones more slowly, ligaments and cartilage slowest of all. Encourage a broad stretch in the muscle belly, and treat sharp or joint centred sensations as a signal to ease off.



KEY POINTS

- Ligament joins bone to bone; tendon joins muscle to bone
- The adult skeleton has 206 living, remodelling bones
- Articular cartilage has no blood vessels and heals very slowly
- Muscle adapts in weeks; tendon and bone take months
- Fascia is real and interesting, but many popular claims are unproven

FEEL IT

Sit with one leg extended and find your Achilles tendon with your fingers. Point and flex your foot slowly and feel the calf muscle and tendon move. Then come into Downward Dog and pedal your heels for five breaths. Note where you feel stretch and describe whether it is in muscle belly, tendon or joint.

CUE IT

“Look for a broad stretch in the belly of the muscle, not a sharp pull near the joint.”

“Bend your knees as much as you need so the stretch stays in the calves.”

“Build intense poses gradually; tendons adapt more slowly than muscles.”

CHECK IT

Which tissue connects muscle to bone?

Answer: Tendon. Tendons connect muscle to bone; ligaments connect bone to bone.

Key terms

Anatomical position	The reference posture for all movement terms: standing, face and feet forward, palms facing forward. Mountain with palms turned out.
Superior and inferior	Toward the head and toward the feet. In Mountain the knees are superior to the ankles.
Medial and lateral	Toward and away from the midline. The big toe is on the medial side of the foot.
Proximal and distal	Closer to or farther from where a limb joins the trunk. In Downward Dog the wrist is distal to the elbow.
Sagittal plane	The plane dividing left from right; forward and backward movements happen in it, as in a Forward Fold.
Frontal (coronal) plane	The plane dividing front from back; side to side movements happen in it, as when the arms lift into Warrior II.
Transverse plane	The plane dividing top from bottom; rotations happen in it, as in a seated twist.
Axis	The imaginary hinge line a movement turns around, always at right angles to the plane of movement. A forward fold turns around a side to side axis through the hips.
Flexion	A movement that usually decreases the angle between two bones, like bending the knees in Chair.
Extension	A movement that usually increases the angle between two bones, like straightening the knees back to Mountain.
Abduction	Moving a limb away from the midline, like lifting the arms out to the sides in Warrior II.
Adduction	Moving a limb toward the midline, like bringing the arms back down to the sides after Warrior II.
Internal and external rotation	Turning the front of a limb toward (internal) or away from (external) the midline. The front thigh in Warrior II is externally rotated.
Lateral flexion	Bending the spine or neck sideways, as in Gate pose or the side tilt of Triangle.
Circumduction	A circular movement combining flexion, abduction, extension and adduction, like arm circles in a warm up.
Dorsiflexion and plantarflexion	Lifting the top of the foot toward the shin (Chair) and pointing the foot (tops of feet down in Cobra).
Inversion and eversion	Tilting the sole of the foot inward and outward; you feel both as the standing foot wobbles in Tree.
Pronation and supination	Turning the palm down (hands on the mat in Downward Dog) and turning the palm up (palms up in Savasana).
Protraction and retraction	Wrapping the shoulder blades forward around the ribs (pressing away in Plank) and drawing them toward the spine (Cobra).

Elevation and depression	Lifting the shoulder blades toward the ears and lowering them away, as when softening a shrug in Chair.
Upward and downward rotation	Turning the shoulder blade so the socket faces up as the arms rise overhead, and back again as they lower.
Horizontal abduction and adduction	With the arm at shoulder height, sweeping it back and out (opening into Warrior II) or across the chest (Eagle arms).

Myths to retire

Warrior II and Triangle must be performed flat, as if between two panes of glass.

The frontal plane is a useful reference, but most pelvises turn slightly toward the floor. Forcing a flat shape can strain knees and the lower back.

Shoulders should always be pinned down and back.

Lifting the arms overhead requires the shoulder blades to rotate upward and elevate a little. Pinning them down fights normal shoulder mechanics.

Twists wring toxins out of the organs.

The liver and kidneys clear waste continuously. There is no evidence that twisting squeezes out toxins, though twists can feel good and improve spinal mobility.

Stretching releases emotions stored in fascia.

Fascia is richly innervated and stretching can feel emotional, but there is no evidence that fascia stores emotions or that poses release them.

Ligaments should be stretched to gain flexibility.

Ligaments stabilise joints and heal slowly. Flexibility work should target muscle, with a broad stretch rather than sharp joint sensation.

Self check

1. What is anatomical position and why do we need it?
2. Name the three cardinal planes and one pose movement in each.
3. What is an axis, and how does it relate to its plane?
4. Describe the front hip in Warrior II using three movement terms.
5. Why can the cue to pin the shoulders down and back be unhelpful when the arms are overhead?
6. What is the difference between a ligament and a tendon?
7. What can you honestly say about fascia to a curious student?
8. A student reports tingling down the leg in a Forward Fold. What do you do?

Model answers

1. Standing upright with face and feet forward and palms facing forward. It gives a single reference so every movement term means the same thing in any posture.
2. Sagittal: hip flexion in a Forward Fold. Frontal: arm abduction into Warrior II. Transverse: spinal rotation in a seated twist.
3. An axis is the hinge line a movement turns around. It always lies at right angles to the plane in which the movement happens.
4. The front hip is flexed, abducted and externally rotated.
5. Overhead reach needs the shoulder blades to rotate upward and elevate slightly. Pinning them down works against that normal movement.
6. A ligament joins bone to bone and guides joint motion; a tendon joins muscle to bone and transmits muscle force.
7. Fascia is connective tissue that wraps and links muscles and organs and is rich in sensory nerves. Claims that it stores emotions or can be quickly remodelled by stretching are not supported by good evidence.
8. Have them ease out of the pose and offer a comfortable alternative. Suggest they see a doctor or physical therapist, without guessing at a diagnosis.